

It should be noted that the base conversion method only works on Integers, passing a float value will raise a `TypeError`.

The python numbers and the `Numbers` module have a vast scope and capabilities that are worth exploring. The expanse of the module and the related arithmetics cannot be contained in a few pages, hence further reading is advised.

Strings

Python comes with a class already built for all String operations called `'str'` with a bunch of handy features that are provided out of the box. At the bare bones, any literal enclosed within a pair of double or single quotes qualifies as a string with the exception of special characters that have to be escaped using a backslash (`'\'`). String literals can span multiple lines if enclosed within a pair of 3 identical single or double quotes (`"""` or `'''`).

In case the string contains a lot of special characters, you may want to consider a raw string that ignores all escape codes. Special characters in a raw string hold no special value, rather they are treated as regular literals with the exception of quotes since an interpreter would treat a quote (`"` or `'`) character as the end of the string, hence it needs escaping. A raw string is prefixed with an `r` or `R` followed by the string in quotes.

```
• >>> a = "abc\txyz"
• >>> print(a)
• abc    xyz          #\t is treated as a tab space
•
• >>> a = r"abc\txyz"
• >>> print(a)
• abc\txyz          #\t is treated as a two character
string
```

Similar to Java, String in python are immutable objects which implies that Strings once created in Python cannot be changed. So string addition doesn't actually add one string to another, rather it creates a new string with an addition of both.

String Slicing

String characters or sub-strings can be accessed using their indexes, an operation which is also called `Slicing`. Python as a language does not support character data type, hence characters are represented as unity